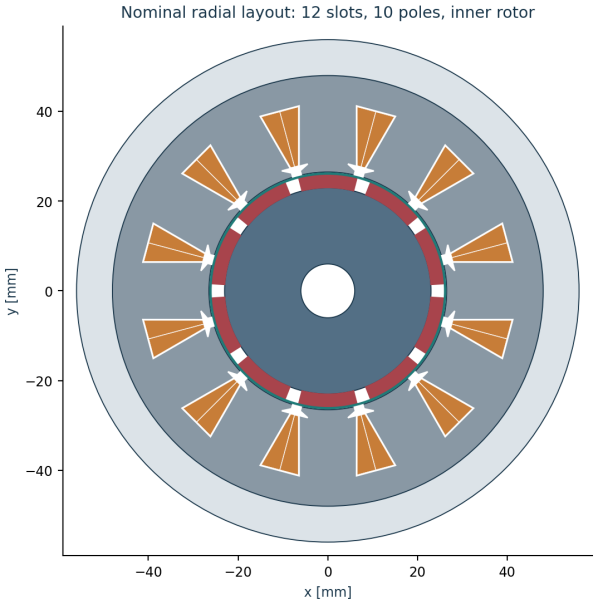
Motor design and Gmsh–Elmer analysis

Space Octopus • 12-slot / 10-pole permanent-magnet motor • 6 November 2026



This report extends the previous Pyleecan/FEMM dimension report with actual three-dimensional geometry, electromagnetic finite elements, insulated-wire thermal calculations, complete motor conduction, and rotor structural calculations. Gmsh prepares the geometry and meshes; Elmer performs the field solves. CalculiX supplies the orthotropic rotor and contact analyses.

The deliverable is a simulation design candidate for independent validation. Supplier magnetic curves, composite strength, adhesive temperature/creep properties, bearing data and manufacturing tolerances remain material design inputs. Numerical completion does not constitute a fabrication or hardware qualification.

Torque finding: at 12.2 Arms and 120°C magnets, the eight-position 3D mean is 4.9001 N·m. After the stated 18 W non-copper loss allowance, the estimated mean shaft torque is 4.6273 N·m against the 4.3651 N·m requirement.

Thermal limit: continuous 22 A hot holding reaches 154.7°C and fails the declared 130°C limit. The refined 18.5 A case reaches 126.4°C. A five-second 22 A pulse from the 12.2 A hot holding field reaches 103.3°C under the stated cooling and material assumptions.

1. Requirement, operating point and interpretation

Quantity	Value / meaning
Actuator torque reference	220 N·m; actuator mechanics are outside this report
Assumed reduction / efficiency	63:1 / 0.80
Actuator / motor speed	10 rpm / 630 rpm
Required motor shaft torque	220/(63×0.80) = 4.365079 N·m
Required motor shaft power	287.979 W at 630 rpm
DC supply	48 V nominal; 42 V minimum
Current ceiling	22 Arms; pulse/voltage/temperature limits apply
Declared winding temperature limit	130°C
Cooling boundary	Distributed barrel-to-sink resistance 0.05 K/W
Sink cases	50°C nominal; 70°C hot boundary
Mechanical screening speed / torque	1,890 rpm / 13.416 N·m; final B outward pressure 2 MPa

The original native model required approximately 10.834 Arms for the nominal target. The additional 12.2 Arms case is a design-margin scenario; its electromagnetic result must be read with the corresponding temperature, rotor position and steel model. A static torque point is not a full-period mean rating.

The 22 A steady thermal calculation tests thermal capacity. It does not establish that 22 A is electrically available at every speed on a 42 V bus. The earlier angle-resolved optional 400 N·m actuator point failed the reserved 42 V voltage criterion and is not reclassified as a pass here.

All dimensions in the sketches/tables are millimetres unless marked otherwise. The motor has an inner rotor: the shaft and magnet-bearing rotor turn inside the stator. The motor contains no second internal motor.

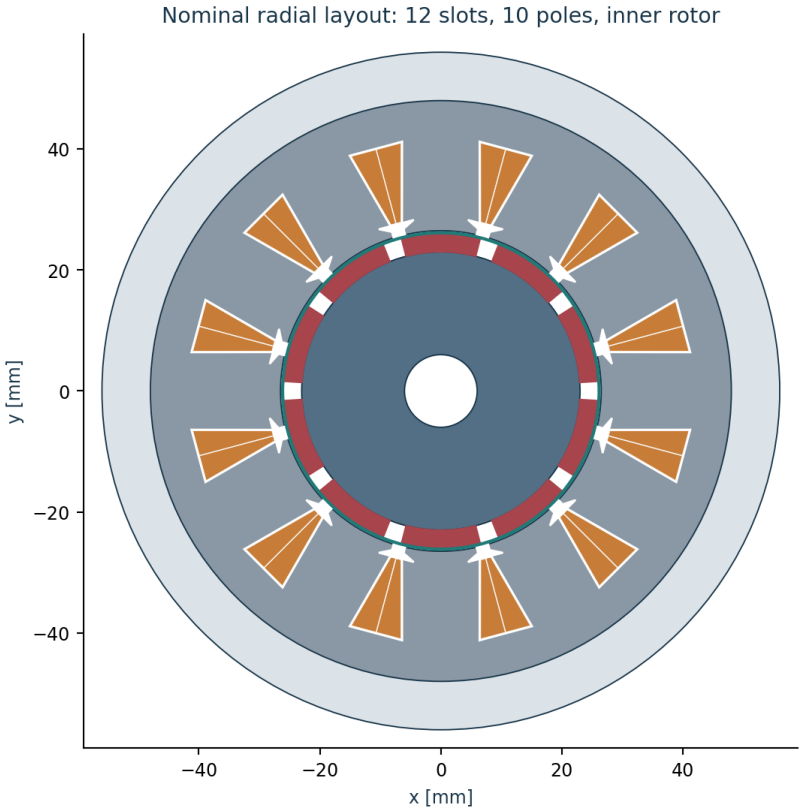
2. Changes from the preceding PDF

Item	Earlier dimensional basis	Declared three-dimensional candidate
Shaft	Basic Ø12 × 130	Ø12 × 155; z = −35 to 120
Coil end geometry	12 mm electrical half-turn allowance; physical route unresolved	Closed stranded-coil envelope with 180° formed end turns
Mean turn length	218 mm	220.174618 mm
Net coil-side envelope	35.75 mm² native half-slot	31.410266 mm² after liner/separator
Magnet bond	Not a separate radial solid	0.05 mm outside rotor iron; magnet radii 23.05–26.05
Outer magnet retention	Proposed 0.20 mm wrap	Wrap radii 26.05–26.25; cold free radial gap 0.25
Magnet axial segmentation	Four segments proposed	Four × 21.175 mm, three × 0.10 mm insulating seams
Housing / end support	Preliminary envelopes	Barrel, two end caps, bearings, collars, nuts and 12 bolts in CAD
Sleeve capture revision B	Revision A lips touch the 0/85 mm core end faces	0.15 mm nominal axial clearance at each retention lip; sleeve z −0.15 to 85.15

Revision A was retained as an analysed comparison. Bonding its sleeve/lip interfaces imposes strong thermal constraints and produces high local stresses. Revision B introduces axial clearance and explicitly evaluates sliding sleeve contact. Its active magnetic solids are unchanged, so the magnetic geometry is shared. Revision B receives a separate thermal and structural check.

The former report remains a historical reference: Motor_Dimensions_and_Analysis.pdf, SHA-256 560495945e37bcd89a2af78bca3e1a77938f5467ca1e12a78cf3cccc54d999f3. This report identifies every new geometric proposal explicitly.

3. Radial dimensions and magnetic stack



Feature	Radius / diameter	Axial extent
Stator outer surface	R48 / Ø96	z 0–85
Stator bore	R26.5 / Ø53	z 0–85
Rotor iron	R6–23 / Ø12–46	z 0–85
Magnet adhesive	R23–23.05	Under each segment
Magnets	R23.05–26.05 / bare Ø52.10	Total magnetic length 84.70
Retention sleeve	R26.05–26.25 / Ø52.50	Revision B z −0.15–85.15
Free radial mechanical gap	26.50−26.25 = 0.250	Excludes thermal growth/runout
Housing barrel	R48–56 / Ø96–112	z 0–85

4. Stator tooth and slot sketch coordinates

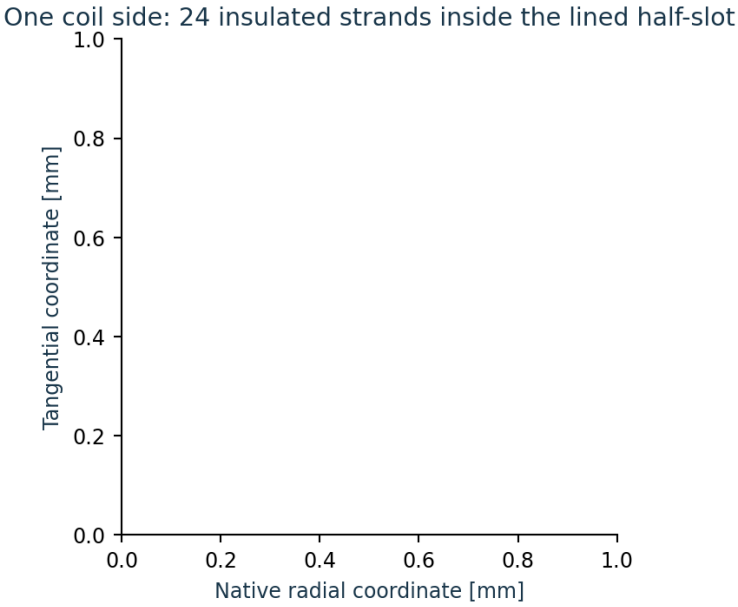
Native point	Radial x [mm]	Tangential y [mm]
Z1	26.481125	-1.000000
Z2	27.481125	-1.000000
Z3	28.481125	-3.000000
Z4	28.481125	-1.000000
Z5	41.481125	-4.500000
Z6	41.481125	4.500000
Z7	28.481125	1.000000
Z8	28.481125	3.000000
Z9	27.481125	1.000000
Z10	26.481125	1.000000

Use straight segments Z1→Z2→…→Z10 and close along the bore arc of radius 26.5 mm through (26.5, 0). Rotate this local slot by 15° for slot 1 and repeat at 30° pitch. This preserves the native SlotW10 tooth tips and slot neck; a rectangular replacement changes the electromagnetic geometry.

Slot parameter	Dimension
W0 / H0	2 / 1
W1 / H1	6 / 1
H2 / W2	13 / 9
Total native slot opening area	77.474832 mm²
Native active winding window	71.500 mm²; 35.750 per side before new insulation
Minimum nominal back iron	6.275502 mm
Outer slot liner / middle separator	0.20 / 0.10 total

The liner and central separator are physical CAD volumes. Their nominal dimensions reduce each active half-slot to the polygon on the next page. No insulation strength, partial-discharge limit or process tolerance is established by geometric fit.

5. Coil-side packing and insulation



Quantity	Value
Turns per coil / strands in hand	12 / 2
Bare conductor diameter	0.950 mm
Assumed radial enamel thickness	0.025 mm
Insulated diameter	1.000 mm
Net half-slot area	31.410266 mm²
Bare copper fraction of net envelope	54.1598%
Insulated-wire area fraction	60.0108%
Minimum wire centre separation	1.02000 mm
Minimum coating-to-envelope clearance	18.660 µm

Twenty-four coated circles fit at nominal dimensions. The small minimum residual clearance is not a manufacturing tolerance allowance. The selected wire must meet the stated maximum coated diameter, and the liner/separator process must preserve the net envelope.

Net polygon vertex	x [mm]	y [mm]
1	28.681125	-0.846724
2	41.281125	-4.239032
3	41.281125	-0.050000
4	28.681125	-0.050000

6. Wire centre coordinates for CAD

Strand	Local x [mm]	Local y [mm]
1	29.779965	-0.605437
2	30.799965	-0.605437
3	31.819965	-0.605437
4	32.839965	-0.605437
5	33.349965	-1.488783
6	33.859965	-0.605437
7	34.369965	-1.488783
8	34.879965	-0.605437
9	35.389965	-1.488783
10	35.899965	-0.605437
11	36.409965	-1.488783
12	36.919965	-2.372129
13	36.919965	-0.605437
14	37.429965	-1.488783
15	37.939965	-0.605437
16	37.939965	-2.372129
17	38.449965	-1.488783
18	38.959965	-0.605437
19	38.959965	-2.372129
20	39.469965	-1.488783
21	39.979965	-0.605437
22	39.979965	-2.372129
23	40.489965	-1.488783
24	40.489965	-3.255475

These are nominal centres for one lined half-slot. Mirror the tangential coordinate for its opposite half, then rotate the slot coordinate frame to the required slot centre. They locate conductor cross-sections; they are not a fabrication routing of all end-turn strands or terminal leads.

7. Winding connections and physical end turns

Coil	Phase	Signed turns	Tooth axis [deg]
1	A	12	0
2	A	-12	30
3	B	-12	60
4	B	12	90
5	C	12	120
6	C	-12	150
7	A	-12	180
8	A	12	210
9	B	12	240
10	B	-12	270
11	C	-12	300
12	C	12	330

The three phases are Y connected, with four series coils and 48 series turns per phase. Signed turns specify the current direction relative to the positive local coil direction. Coil side T0 in slot s returns through the mirrored T1 side in slot s−1. The table retains the native Pyleecan connection matrix.

End-turn quantity	Value
Closed mean turn length	220.174618 mm
Active straight-leg contribution	2 × 85 = 170 mm
Mean total end-turn contribution	50.174618 mm
Nominal cold phase resistance	0.128522351 Ω at 20°C
Copper temperature coefficient	0.00393 K^{-1}
Physical axial end-turn overhang	Approximately 10.636 mm from the exact coil-side profile

The CAD coils are connected homogenized stranded-current solids. The end-turn geometry is a 180° revolution of the lined coil-side polygon about its tooth radial axis at each stack end. The electromagnetic source follows that closed loop; it is not a shorted solid copper ring.

An independent CAD volume check found about 0.123% difference between kernel quadrature and the analytical swept-volume value for each curved coil. Electrical resistance uses the conservative analytical 220.174618 mm turn length. This small geometry/integration difference is disclosed rather than hidden in an exact-volume claim.

Conductor loading	Value
Two-strand copper area per turn	1.417644 mm^2
Copper current density at 12.2 / 13 / 22 Arms	8.606 / 9.170 / 15.519 A/mm^2
Worst holding phase current density	√2 times the corresponding dq RMS value

8. Magnets, sleeve, adhesive and capture clearances

Feature	Nominal definition
Pole pitch / arc	36° / 28.8°
Pole 1 angular extent / centre	3.6°–32.4° / 18°
Magnetization	Alternating outward/inward radial direction
Segments per pole / total	4 / 40
Segment axial length / seam	21.175 / 0.100
Segment z intervals	0–21.175; 21.275–42.450; 42.550–63.725; 63.825–85.000
Inner bond thickness	0.050 radial
Sleeve radial thickness	0.200
Revision B sleeve z interval	−0.150 to 85.150
Revision B front / rear retention lips	−0.650 to −0.150 / 85.150 to 85.650
Lip radial interval	22.500 to 26.250
Cold axial core-to-lip clearance	0.150 each side

The radial sleeve consumes mechanical clearance but is treated as approximately nonmagnetic. The bond moves the magnet radially outward by 0.05 mm. The total nonmagnetic radial path from rotor iron to stator remains approximately 0.50 mm because the inner bond replaces an equal amount of external bare-magnet gap.

Revision B axial clearance is a proposed change derived from the hot restraint study. A stress-free 20°C steel stack growing to 150°C increases its 85 mm length by approximately 0.1326 mm, or 0.0663 mm per end about the midplane. The stated 0.15 mm nominal clearance leaves approximately 0.0837 mm per end before tolerances and differential sleeve effects.

9. Shaft, bearings and enclosure dimensions

Part / feature	Basic dimensions and locations
Shaft	Ø12 × 155; z −35 to 120
Equivalent 6201 bearings	Ø12 ID × Ø32 OD × 10; centres z −14 and 99
Bearing seat intervals	Front z −19…−9; rear z 94…104
Bearing support span	113
Barrel	Ø96 ID / Ø112 OD × 85
End caps	Front z −20…0; rear z 85…105
End-cap plate thickness	6; plates z −20…−14 and 99…105
Bearing boss / shoulder opening	Basic Ø44 boss / Ø28 shoulder opening
Inner collars	Ø12/18 × 4; z −9…−5 and 90…94
Locknut envelopes	M12×1 equivalent, Ø19 × 3; z −22…−19 and 104…107
Cap screws	Six M4×30 per end; PCD104; cap clearance Ø4.5
Barrel screw engagement	10 mm at each end; Ø4 thread-equivalent cylinders

Bearings, collars, nuts and fasteners are dimensional envelopes in CAD. Bearing internal geometry, preload, journal fits, screw thread helices, rotor-to-shaft joint capacity and output coupling interfaces are not selected by the electromagnetic solver. These must be assigned for a manufactured assembly and for a final bearing/contact model.

10. Verified Pyleecan–Elmer coupling

The installed Pyleecan source is kept unchanged. The stock MagElmer mesh-conversion method is used; this project supplies the missing physical field/flux extraction through a motor-specific adapter. That adapter runs inside native Simu1, integrates depth exactly once, returns actual B fields and phase fluxes, and feeds native ForceMT.

Check	Executed result
Native Pyleecan API angular positions	24
Elmer mean torque [N·m]	4.440
Reference FEMM mean torque [N·m]	4.427214318
Mean-torque difference	Approximately 0.290% for 24 samples
Same-mesh vector-potential comparison	Linear and nonlinear Elmer versus independent scikit-fem: about 0.0117% relative error
Force coupling	Native Pyleecan ForceMT executed with returned magnetic field
Depth normalization	Project flux/torque divided by stack length before native integration

The earlier 12-position torque-ripple result aliased the waveform and was replaced by the 24-position run. The 24-position peak-to-peak torque ripple is about 0.654 N·m. Mean torque agreement alone does not establish ripple convergence or all high-frequency losses.

This is an explicit project repair of an incomplete upstream interface, not a claim that the unmodified stock MagElmer extractor supplies all of these quantities. The adapter and readback are included with the reproducible study.

11. Three-dimensional magnetic mesh and constitutive law

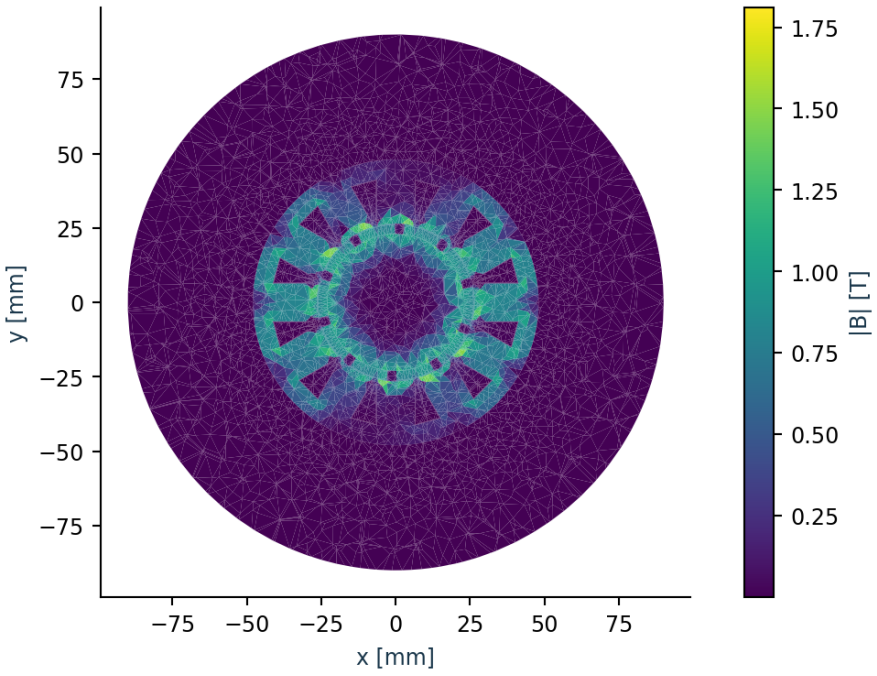
Mesh	Nodes	Tetrahedra	Near-gap target [mm]
coarse	65307	367755	1.200
fine	141627	803193	0.700
segmented	65990	372724	1.200

The axial-half domain starts at the 42.5 mm midplane. Magnetic normal flux is zero on that symmetry plane; total torque and phase flux integrals are multiplied by two. The far cylinder radius is 90 mm and its outer axial plane is z = 155 mm. The complete closed coil end region is present. The nonmagnetic housing, liner, bond and wrap are replaced by air in static magnetics; exterior ferromagnetic fastener effects are a separate unresolved material scenario.

The STEP rotor is at the nominal CAD reference, with pole 1 centred at 18°. The first magnetic case adds 50.999972° mechanical rotation to that rotor while keeping stator/coils fixed. Its phase currents are [0, +√(3/2) Iq, −√(3/2) Iq]. Other sweep cases use electrical angle 5(θ−50.999972°). Preserve this reference when recreating the cases in Ansys; CAD zero and the first solved rotor position are different.

The linear check uses relative permeabilities 1500 in the lamination plane and 19.75 axially. The nonlinear model uses the bundled M270-35A B–H curve and 95% steel fill, mixing parallel permeability and series axial reluctivity. The supplied curve ends at 1.8 T; higher-field extrapolation remains an unqualified material assumption.

A boundary-rooted spanning-tree gauge removes every active gradient mode and is checked by graph rank. The native interior forest left one mode in this mesh. Failed/nonphysical configurations are excluded. Direct and iterative comparisons, mesh refinement, finite field checks and actual solver residuals are recorded; an Elmer process merely exiting successfully is insufficient.



12. Executed three-dimensional magnetic results

Mesh / case	Iq [Arms]	Magnet [°C]	Torque [N·m]	ψd,rms [Wb]	Peak |B| [T]
segmented/design12p2_80C	12.200	80.0	4.988276	0.0272065	3.0905
segmented/fault_minus55d_150C	0.000	150.0	0.005064	-0.0198218	3.0133
segmented/peak22_150C	22.000	150.0	8.808578	0.0266198	3.0309
segmented/design12p2_120C	12.200	120.0	4.927990	0.0268765	3.0529
coarse/linear_rooted_cholmod	10.814	80.0	4.411918	0.0268723	2.1799
coarse/nonlinear_tensor_newton	10.814	80.0	4.472416	0.0272560	2.0071
fine/linear_bicg	10.814	80.0	4.382280	0.0268217	2.4691
fine/nonlinear_tensor_newton	10.814	80.0	4.444000	0.0272112	1.7773

The accepted torque is Elmer’s Component variational torque, multiplied by two for axial symmetry. It is cross-checked by independently integrating the air Maxwell tensor against a piecewise-linear nodal virtual rotation. The legacy Torque Group readout and a separately integrated narrow continuous weighting function disagree with that mesh-consistent result and are excluded from rating conclusions.

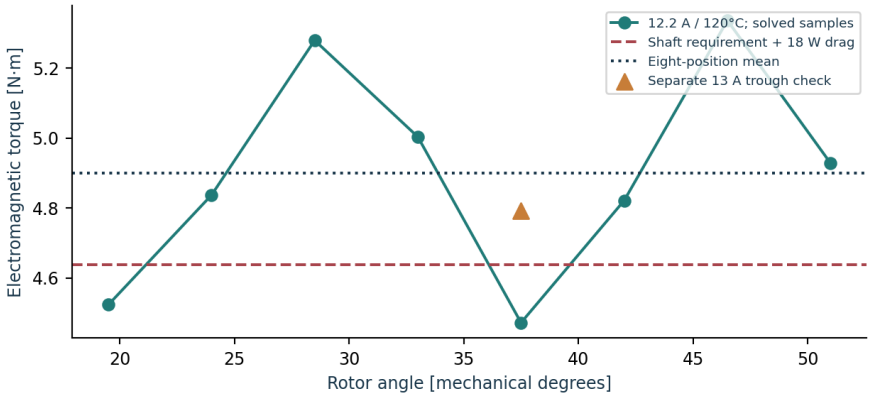
Torque method / check	Linear coarse result [N·m]
Component variational torque	4.411918
Independent discrete air Maxwell tensor	4.411387
Fixed-current energy derivative, ±0.25°	4.410494
Fixed-current energy derivative, ±0.50°	4.407909

For linear tetrahedral Nedelec elements, A inside a cell is reconstructed as A at the centroid plus ½ B×(r−rcentroid). Coil flux is then integrated with 64-point Duffy quadrature along the actual curved current direction. The 27-versus-64-point change is recorded to separate end-turn integration error from magnetic mesh error.

With the accepted component method, coarse-to-fine linear torque changes 0.672% (4.411918→4.382280 N·m); nonlinear torque changes 0.635% (4.472416→4.444000 N·m). The fine nonlinear ψd,rms is 0.02721123 Wb. The independent discrete Maxwell torque differs from the component result by less than 0.02% in the principal comparison cases. Local sharp-corner |B| maxima are less converged than integral torque, especially near the axial magnet seams.

The approximately 3 T global maxima in the segmented cases occur in tiny air regions near the inner bond/seam edges, not as a uniform steel induction. At the 12.2 A/120°C reference position, the maximum homogenized steel |B| is about 1.686 T and its volume-weighted p99 is 1.303 T. Intrinsic lamination BFe can exceed bulk B because of fill. The near-zero d-axis fault torque is not a useful relative-error benchmark; its extraction difference is about 0.0016 N·m in absolute terms.

12A. Angular mean, shaft torque and actuator requirement



Rotor angle [deg]	3D torque [N·m]	ψd,rms [Wb]
51.000	4.927990	0.0268765
46.500	5.336322	0.0268698
42.000	4.821576	0.0269097
37.500	4.471982	0.0268176
33.000	5.003394	0.0270008
28.500	5.279470	0.0268859
24.000	4.836896	0.0269332
19.500	4.523192	0.0269530

Quantity	Executed result
Current / magnet temperature	12.2 Arms / 120°C
Repeat interval / samples	36° mechanical / eight positions at 4.5°
Mean electromagnetic torque	4.900103 N·m
Sampled torque extrema	4.471982 to 5.336322 N·m
Four-versus-eight mean change	0.054%
Non-copper shaft-drag allowance	18 W / 65.97345 rad/s = 0.272837 N·m
Estimated mean shaft torque	4.627266 N·m
Required shaft torque	4.365079 N·m
Estimated mean actuator torque	233.214 N·m at 63:1 and 80% efficiency
Mean shaft-torque margin	6.01%

The eight-position sampling spans the native 36° repeat interval. Applied to the existing 24-position 2D data, the same eight positions change the mean by approximately 0.022%. That is a sampling check using the reference waveform; sampled 3D extrema still do not prove ripple convergence. The four-versus-eight 3D mean comparison is reported above.

Electromagnetic torque is converted to an estimated shaft value by deducting the entire stated 18 W non-copper loss envelope at 630 rpm. Copper loss affects voltage, efficiency and temperature but is not subtracted again as shaft drag. Replace the allowance with actual core, magnet, bearing and windage losses when supplier data and measurements are available. Mean torque does not ensure the minimum instantaneous torque without controller ripple compensation.

12B. Flux-derived voltage and hot-duty efficiency

Quantity	Calculated value
Resistance temperature upper bound	82.539°C winding hotspot
Phase resistance upper bound	0.16011012 Ω
Maximum sampled-harmonic dq voltage circle	16.2246 V phase peak
Maximum phase-voltage span	28.0895 V
42 V bus reserved ceilings	21.824 V phase circle / 37.800 V span
Moving copper loss	71.074 W
Additional motor loss allowance	18.000 W
Estimated shaft power at 630 rpm	305.277 W
Estimated motor electrical input	394.351 W
Estimated motor efficiency	77.41%
Estimated 48 V DC current	8.216 A, before inverter loss

The voltage estimate interpolates the eight-position 3D dq flux waveform and differentiates it with rotor angle. With RMS dq quantities, $V_d = R I_d + \omega_m(d\psi_d/d\theta - 5\psi_q)$, $V_q = R I_q + \omega_m(d\psi_q/d\theta + 5\psi_d)$. Three phase voltages are reconstructed and their span is compared with the reserved bus voltage. The resistance uses the full winding hotspot as a conservative phase-temperature bound.

Only harmonics resolved by the eight samples are represented; higher space harmonics and switching components remain outside this new estimate. The earlier full-period FEMM/drive work remains relevant. These values are a nominal-duty model check, not a controller or inverter qualification. The estimated efficiency includes the new hot copper loss and 18 W motor loss allowance; it excludes inverter power loss.

12C. Torque trough and current-control headroom

Check	Result
Selected lowest-torque sampled rotor angle	37.500°
12.2 A electromagnetic / estimated shaft torque	4.471982 / 4.199145 N·m
13 A verified electromagnetic / estimated shaft torque	4.792408 / 4.519571 N·m
Required motor shaft torque	4.365079 N·m
13 A sampled-trough shaft margin	3.54%
13 A steady hot holding winding	96.137°C; refined low-resin/poor-contact model
13 A moving winding / magnet maxima	84.188 / 121.313°C
22 A, 5 s pulse from 13 A holding field	105.964°C winding
Linearized compensation current range	10.601 to 12.693 Arms
Linearized compensation voltage circle	17.522 V peak; reserved ceiling 21.824 V

An additional 13 A trough solve at 150°C magnets gives 4.752052 N·m electromagnetic and 4.479215 N·m estimated shaft torque, a 2.61% margin. This magnet temperature is above the 121.3°C moving thermal result and adds a temperature sensitivity check.

Constant 12.2 A meets the mean torque criterion under the stated assumptions, but the sampled trough after the 18 W drag allowance falls below the shaft requirement. A separate nonlinear 13 A solve at the lowest sampled position tests available headroom. A controller can use an angle/current lookup and feedback to compensate ripple; the table reports a linearized feasibility profile, not an implemented controller.

The profile estimates Iq(θ) by scaling the sampled torque curve, then includes the resulting q-flux derivative in voltage. Cogging is not strictly proportional to current, and current-dependent cross-saturation is only approximated. Exact continuum extrema, sensor/commutation errors, dynamic current tracking and inverter limits require controller calibration. The constant 13 A thermal cases conservatively bound the heating of a profile that remains below 13 A.

Mean-duty sensitivity at 12.2 A	Threshold / scope
Maximum non-copper loss compatible with 220 N·m	35.297 W at 630 rpm
Extra loss budget beyond the assumed 18 W	17.297 W
Minimum gearbox efficiency for this mean shaft torque	75.47% at 63:1
Approximate Br20 floor if torque scaled linearly	0.9938 T; first-order sensitivity only

The 6% mean torque margin is not a material or manufacturing safety factor. Supplier remanence variation, actual gap, loss and gearbox efficiency can consume it. The remanence threshold is an algebraic first-order sensitivity, not another finite-element validation.

13. Electrical use, demagnetization and losses

Model input / inherited result	Value / scope
Magnet Br at 20°C	1.05 T
Br temperature coefficient	−0.0003 K^{-1}
Magnet recoil μr	1.05
Magnet electrical resistivity	0.9 μΩ·m at 20°C; assumed +0.0003 K^{-1}
Phase resistance at 20°C	0.128522351 Ω for new coil length
Phase resistance at 100 / 130°C	0.16892978 / 0.18408256 Ω
Native nominal phase voltage peak	15.734 V at the former nominal operating point
42 V reserved phase-voltage ceiling	21.824 V peak in the inherited drive criterion
Inherited nonlinear fault current / torque	55.27 Arms vector envelope / 13.416 N·m
Inherited reverse-field knee requirement	About 852 kA/m for the prior cold 1,890 rpm nonlinear fault case

A linear recoil magnet model cannot prove irreversible demagnetization resistance. The negative field must be checked against the selected supplier’s temperature-dependent intrinsic curve and knee, including segment-edge fields and tolerances. Br alone does not specify coercivity.

New segmented 3D case	Maximum reverse H [kA/m]	Volume-weighted p99 [kA/m]
design12p2_80C	422.802	336.032
fault_minus55d_150C	711.916	476.501
peak22_150C	431.326	337.275
design12p2_120C	417.896	332.083

The −55.27 A d-axis, 150°C magnet case requires approximately 712 kA/m at its largest sampled local reverse field. This is a directional fault scenario, not an exhaustive current-angle demagnetization envelope. The inherited cold maximum of approximately 852 kA/m remains a separate requirement; neither result supplies the missing intrinsic magnet curve.

The preceding study already ran native winding, core-loss, magnet-loss and force models, full-period FEMM field histories, direct AC wire impedance, DC-biased harmonic magnet response and ideal-switch drive spectra. Steel loss coefficients remain calibrated to a generic 3 W/kg point, not supplier data. The new thermal campaign deliberately uses a stated additional motor-loss envelope rather than claiming those generic coefficients are measured losses.

14. Three-dimensional magnet segmentation and AC response

Mesh	Segments	f [Hz]	Field axis	Pole loss [W] at 10 mT peak
rooted_coarse	1	126	x	0.00123717
rooted_coarse	1	126	y	0.00009175
rooted_coarse	1	20000	x	30.90838684
rooted_coarse	1	20000	y	2.31014604
rooted_coarse	1	630	x	0.03092895
rooted_coarse	1	630	y	0.00229382
rooted_coarse	4	126	x	0.00086415
rooted_coarse	4	126	y	0.00008422
rooted_coarse	4	20000	x	21.59503466
rooted_coarse	4	20000	y	2.12014729
rooted_coarse	4	630	x	0.02160358
rooted_coarse	4	630	y	0.00210552
rooted_fine	1	126	x	0.00123316
rooted_fine	1	126	y	0.00008662
rooted_fine	1	20000	x	30.78416488
rooted_fine	1	20000	y	2.18001168
rooted_fine	1	40000	x	119.92452079
rooted_fine	1	40000	y	8.69242806
rooted_fine	1	80000	x	438.39037821
rooted_fine	1	80000	y	34.35897052
rooted_fine	4	126	x	0.00086131
rooted_fine	4	126	y	0.00007991
rooted_fine	4	20000	x	21.51000121
rooted_fine	4	20000	y	2.01099517
rooted_fine	4	40000	x	83.89039661
rooted_fine	4	40000	y	8.01494314
rooted_fine	4	80000	x	307.78901661
rooted_fine	4	80000	y	31.62939164

These are three-dimensional harmonic A–V conductor solves with electrical isolation of each segment, finite conductivity, closed return currents and internal conductive reaction. A uniform boundary induction is clamped on one pole; surrounding motor steel/air reaction and the actual rotating/PWM boundary spectrum are absent. Multiply by ten poles only for a genuinely identical excitation on every pole. Loss scales with the square of boundary amplitude in this linear test.

A separate low-frequency conduction formulation was solved with Elmer StatCurrentSolver and independently with scikit-fem on the same 3D mesh. Potential differences were below 0.00001% and spurious reference-node current was negligible. Four axial segments reduce the tested radial-field low-frequency loss by approximately 30%; tangential-field reduction is much smaller. A $1/N^2$ segmentation rule is not valid for this magnet geometry.

14A. Recorded PWM spectra and segmentation sensitivity

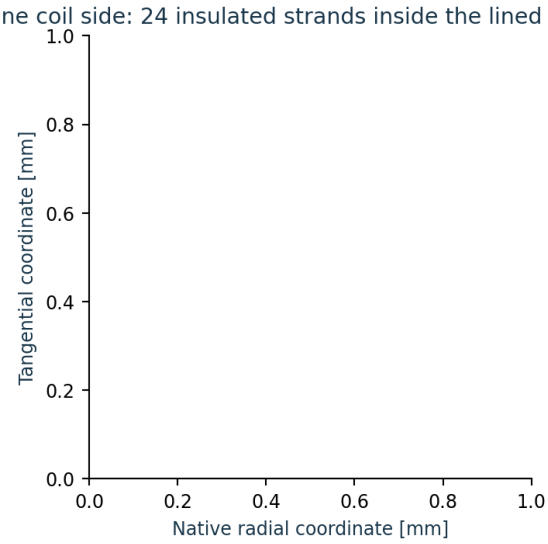
f [kHz]	Four/one segment loss ratio, radial	Four/one segment loss ratio, tangential
20	0.698736	0.922470
40	0.699527	0.922060
80	0.702089	0.920557

Recorded current waveform	2D continuous reference [W]	3D shape sensitivity [W]
PWM_42_0_100C	0.086861	0.060773–0.080083
PWM_42_1e-06_100C	0.084728	0.059274–0.078120
PWM_42_1e-06_130C	0.085407	0.059748–0.078746
PWM_48_1e-06_100C	0.111742	0.078186–0.103020
PWM_150C_substeps40	0.091705	0.064151–0.084556
PWM_150C_substeps80	0.085895	0.060091–0.079197
PWM_20C_substeps40	0.093751	0.065574–0.086446
PWM_20C_substeps80	0.085158	0.059574–0.078518

The recorded dq current spectra are applied to the earlier full-motor DC-biased Hermitian FEMM response at 20/40/80 kHz. The new Elmer isolated-pole four/one-segment ratios are then used as directional shape sensitivities at those frequencies. They predict roughly 0.06–0.10 W magnet PWM loss for those recorded waveforms, well below the stated 3.6 W rotor heat allocation in the 18 W moving-loss scenario.

This is a quantified transfer sensitivity, not a complete 3D motor PWM loss bound. The actual motor fields are nonuniform; the old drive waveforms/bias are retained; response temperature is 80°C. Approximately 25–41% of the current variance lies outside 20–80 kHz, where the original response uses a low-frequency clamp or high-frequency continuation. The new 13 A compensation controller is not re-simulated here. Winding proximity, steel PWM loss and inverter power remain separate uncertainties.

15. Thermal conductivity from 24 actual insulated strands



Resin k [W/m·K]	Conservative transverse k	Dirichlet tensor eigenvalues
0.400	0.991185	1.078015, 1.296595
0.200	0.584125	0.670977, 0.906967

Copper is 390 W/m·K and enamel is 0.15 W/m·K. The actual 24-wire cross-section is meshed, including every enamel ring. Dirichlet and Neumann apparent-conductivity experiments are run in two directions. The minimum Neumann-tensor eigenvalue is used as a conservative transverse scalar in the motor. The along-wire conductivity is approximately 211.3 W/m·K from actual copper/enamel/resin fractions.

The winding conductivity tensor follows the curved end turn: $K = k_t I + (k_a - k_t)\, t \otimes t$. A fixed global axial conductivity would incorrectly conduct through the curved end region. These values remain conditional on impregnation quality and the assumed resin/enamel properties; they are not measurements of a manufactured winding.

16. Full-motor thermal model and cooling requirements

Thermal feature	Declared value / treatment
Domain	Axial half of complete declared motor solids
Vacuum-facing surfaces	Adiabatic; no internal gas convection or radiation
External sink	Full Ø112 × 85 barrel only; 0.05 K/W distributed Robin boundary
Equivalent barrel coefficient	668.718 W/m²·K = 1/(0.05 × 2π × 0.056 × 0.085); equivalent sink coupling
Stator-to-housing interface	50 µm equivalent layer; k = thickness × contact conductance
Contact conductance cases	5,000 and 1,000 W/m²·K
Stator / rotor laminated k	20 in-plane / 1.5 axial W/m·K
Aluminium k	167 W/m·K
Liner / bond / wrap k	0.2 / 0.2 / 0.3 W/m·K
Equivalent bearing k	0.5 W/m·K; no measured bearing resistance
Copper source	Local temperature-dependent I^2R, except explicitly fixed-hot pulse bound

All coil end-turn heat and heat capacity are present. Moving operation applies phase RMS heating. Holding uses instantaneous balanced phase currents. The initial angle-zero pattern [0, +√(3/2) Iq, −√(3/2) Iq] was extended by a 0°/15°/30°/45°/60° electrical-angle sweep. The hottest selected position is 30°, with [−Iq/√2, +√2 Iq, −Iq/√2], concentrating heat in one phase. Every pattern has the same total current square, 3 Iq².

Additional motor losses are distributed 80% into stator iron and 20% into rotor iron plus magnets, proportional to their volumes. The moving design cases use 18 W; the 22 A moving screening case uses 25 W; hot holding/pulse cases use 2 W. These are stated heat-load scenarios, not a measured or supplier-calibrated partition.

The 50 µm thermal contact layer replaces only the outermost stator steel in the FE conduction model. It is an equivalent resistance, not an extra mechanical CAD gap. Exterior bolts are omitted from thermal meshing; cap/barrel contact is idealized. Bearing and interface data must be replaced with measured or supplier values for a final thermal qualification.

The basic Ø96 stator/barrel dimensions do not define an interference fit. With generic aluminium/steel expansion coefficients 23/12 µm/m·K, common 50 K heating creates 26.4 µm radial differential expansion at R48. A selected fit, bond or retained interface must preserve the assumed conductance as temperature changes. Fit tolerances and retention are not supplied by thermal conductivity alone.

Thermal transient steel density is the full-material 7650 kg/m³; the nominal CAD mass table applies 95% fill to laminated steel. This small heat-capacity difference is an input uncertainty. Homogenized coil temperature does not resolve a defective impregnation void or an individual poorly cooled strand.

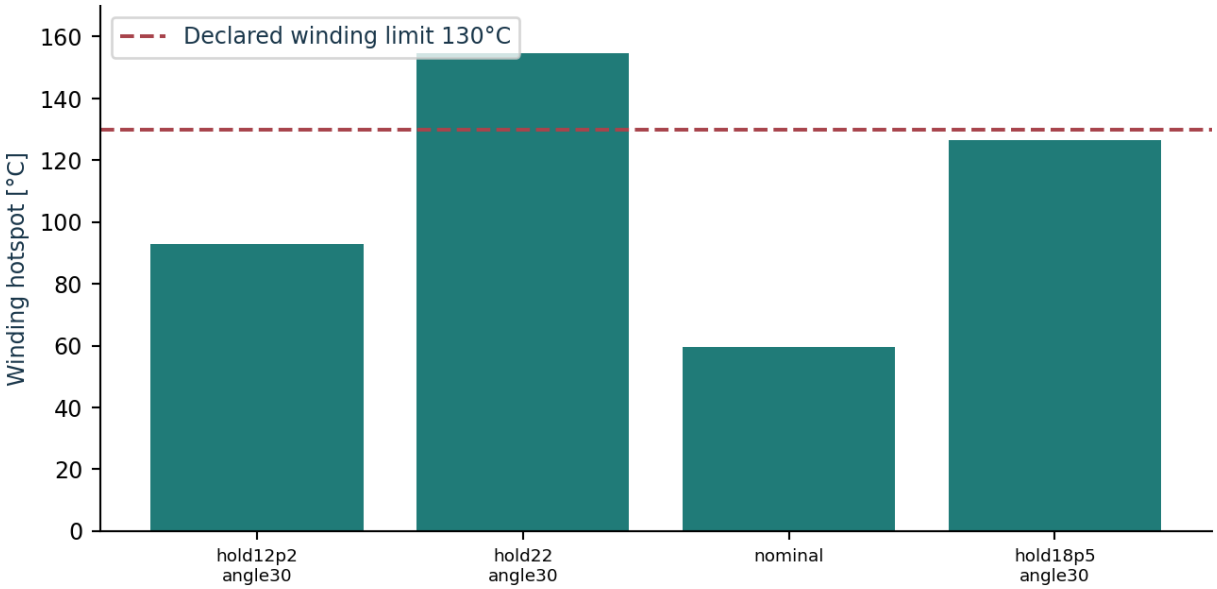
16A. Thermal material properties for independent recreation

Material	Density [kg/m³]	Heat capacity [J/kg·K]	k [W/m·K]
Copper	8960	385	390
Enamel	1200	1400	0.15
Resin	1200	1100	0.4 or 0.2
Laminated stator/rotor steel	7650	460	20 x/y; 1.5 z
Shaft, collars, locknuts	7650	460	45
SmCo magnet	8300	350	10
Aluminium housing/caps	2700	900	167
Liner / wedge / bond	1200	1100	0.2
Sleeve / lips	1600	1000	0.3
Equivalent bearing	5000	460	0.5
Equivalent stator contact	1200	1000	0.25 or 0.05

Impregnated winding property	Value
Homogenized density	5402.7972 kg/m³
Homogenized heat capacity	461.6960 J/kg·K
Copper / enamel / resin volume fractions	0.5415976 / 0.0585105 / 0.3998919
Along-wire conductivity	211.31–211.39 W/m·K; constituent mixture
Conservative transverse conductivity	0.584125 (low resin) / 0.991185 (nominal resin) W/m·K

All values are declared generic inputs. Winding k follows the curved wire direction as described in section 15. The stator contact conductivity is conductance × 50 µm; its density and heat capacity are numerical equivalents. Replace these properties with selected-material data before using the model as a hardware qualification.

17. Steady temperatures and mesh sensitivity



Mesh / case	Iq	Sink [°C]	Winding max [°C]	Magnet max [°C]	Cu loss [W]
coarse/continuous22A	22.000	70	111.148	145.865	248.695
coarse/design12p2_hold70	12.200	70	89.930	78.734	71.964
coarse/design12p2_moving70	12.200	70	82.539	119.078	71.074
coarse/holding70	10.834	70	81.818	77.910	55.741
coarse/holding70_low_resin	10.834	70	83.288	77.917	55.821
coarse/holding70_poor_contact	10.834	70	85.590	77.926	56.190
coarse/moving70	10.834	70	80.110	118.288	55.688
coarse/nominal	10.834	50	59.536	98.101	52.045
fine/continuous22A	22.000	70	111.318	148.374	248.752
fine/holding70_low_resin	10.834	70	83.197	78.116	55.814
fine/nominal	10.834	50	59.576	99.907	52.048
revisionB/design12p2_hold70	12.200	70	89.884	78.734	71.964
revisionB/hold12p2_angle15	12.200	70	91.833	78.734	71.969
revisionB/hold12p2_angle30	12.200	70	92.945	78.734	71.970
revisionB/hold12p2_angle45	12.200	70	91.869	78.734	71.962
revisionB/hold12p2_angle60	12.200	70	89.827	78.733	71.952
revisionB/hold18.5_angle30	18.500	70	126.584	84.085	176.420
revisionB/hold19.0_angle30	19.000	70	130.124	84.641	187.290
revisionB/hold22_angle30	22.000	70	154.732	88.483	262.277
revisionB/nominal	10.834	50	59.525	98.100	52.046
revisionB_fine/hold12p2_angle30	12.200	70	92.870	78.924	71.959
revisionB_fine/hold13_angle30	13.000	70	96.137	79.450	82.234
revisionB_fine/hold18p5_angle30	18.500	70	126.387	84.272	176.388

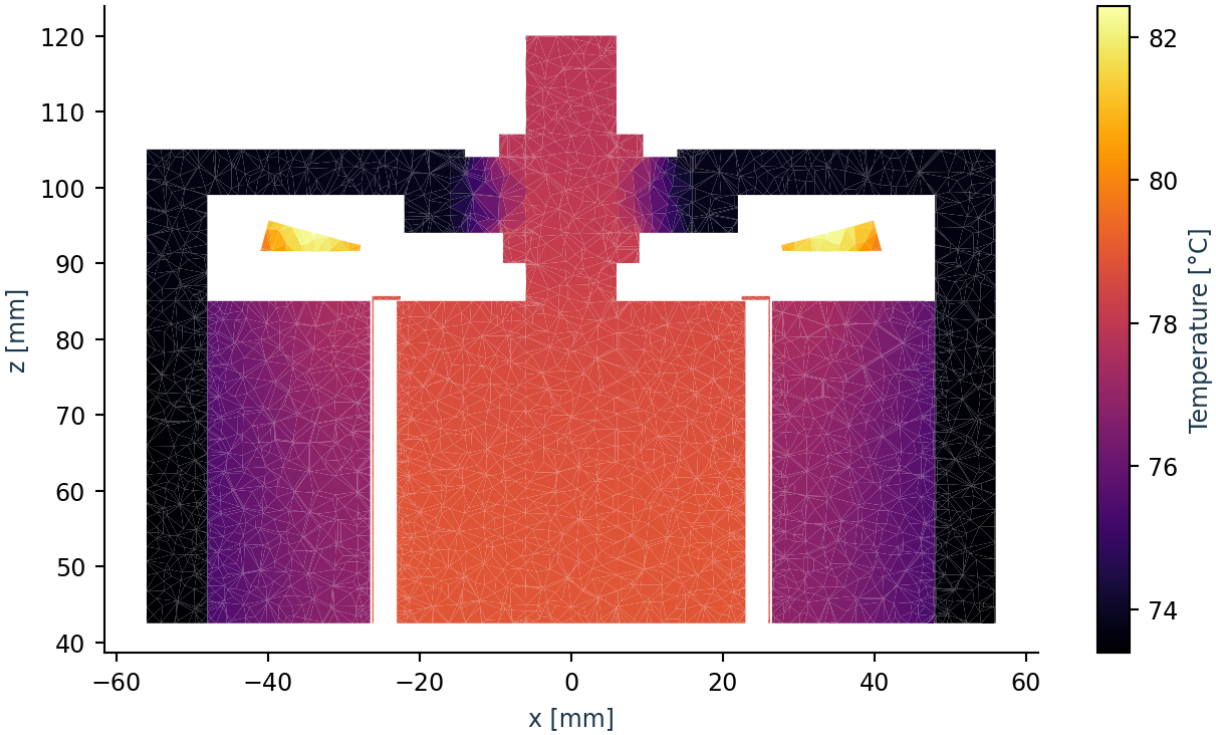
Mesh / case	Iq	Sink [°C]	Winding max [°C]	Magnet max [°C]	Cu loss [W]
revisionB_fine/moving13_hot70	13.000	70	84.188	121.313	81.041

The original coarse and refined full-motor meshes contain 99,278 / 177,525 nodes and 459,445 / 870,663 tetrahedra. Refinement changes the nominal winding hotspot by about 0.04°C, the low-resin hot holding point by 0.09°C, and the 22 A steady thermal case by 0.17°C. These changes quantify numerical sensitivity for the declared material/contact/cooling assumptions.

New holding limit: with a 70°C sink, 0.2 W/m·K resin, 1000 W/m²·K interface conductance and the 30° holding phase pattern, 22 A reaches 154.732°C and fails the 130°C criterion. At 19.0 A the coarse result is 130.124°C. Select 18.5 A as a conservative analysed continuous holding point: 126.584°C coarse, 126.387°C refined. This is conditional on the specified sink and impregnation, not a universal motor current rating.

Revision B’s refined model contains 177,964 nodes and 869,688 tetrahedra. At 12.2 A in the same hot holding scenario, coarse/refined winding temperatures are 92.945/92.870°C, a change of 0.075°C. The 18.5 A change is 0.198°C.

18. End-turn temperature and five-second current pulse



Time steps	Δt [s]	Winding max [°C]	End-turn max [°C]	Imposed Cu [W]
20	0.250	91.681656	90.217131	267.288
100	0.050	91.686162	90.221098	267.288

The table above is the temporal-refinement check for the earlier angle-zero holding case: 20 versus 100 BDF2 time steps change the winding result by 0.0045°C. It is retained as a numerical check, not the worst holding result.

Final 30° holding pulse	Winding max [°C]	End-turn max [°C]	Imposed Cu [W]
revisionB	103.427400	100.424830	267.328
revisionB_fine	103.265990	100.109290	267.289

The final pulse starts from the converged 12.2 A, 70°C-sink, low-resin, poor-interface field at the hotter 30° holding position. Apply 22 A for 5 s with 0.05 s steps and heat based on 130°C copper resistance. The refined winding/end-turn maxima are 103.266/100.109°C; the coarse winding result is 103.427°C. The imposed refined copper heat is 267.289 W. This heat-input bound is conservative while the winding remains below 130°C; actual strand temperatures, heat capacities and cooling must still be qualified.

19. Rotor loading, material model and mesh

Structural input	Value / scope
Full rotor domain	Shaft, rotor iron, 40 magnets, 40 bond pads, sleeve and retention lips
Mesh baseline	228,336 tetrahedra; 364,694 nodes for the A quadratic model
Thermal state	Uniform 150°C rotor from stress-free 20°C assembly
Speed / torque	1,890 rpm / 13.416 N·m
A historical outward pressure	0.591525 MPa from earlier 2D fault screening
B final outward pressure	2.000 MPa, simultaneously on all magnet outer faces
B pressure selection	3D sampled outer-gap $\|B\|_{max}$ 1.817 T → $\|B\|^2/(2\mu_0)$ 1.314 MPa; round up to 2 MPa
Bearings	On-axis mean journal translation springs; 5 MN/m per bearing
Shaft / magnet E	210 / 150 GPa; generic input values
Sleeve E radial / hoop / axial	10 / 100 / 10 GPa; orthotropic cylindrical axes
Sleeve CTE radial / hoop / axial	25 / 0 / 0 µm/m·K
Bond reference law	E 2.6 MPa, ν 0.3; G 1 MPa
Contact sensitivity	Frictionless unilateral sleeve; compressible neo-Hookean bond

The final bearings act on mean journal translations through on-axis reference springs, with 5 MN/m per bearing in each radial direction. They exert no torsional restraint. The output constraint removes mean tip rotation only; the front journal fixes mean axial rigid motion. Magnet-face load integration preserves zero resultant force and exactly 13.416 N·m in the undeformed geometry. Final contact uses linear tetrahedra; the original quadratic geometry benchmark remains a separate reference.

The B sleeve's floating axial reference follows the rotor core's mean midplane translation. Holding the sleeve at zero absolute axial displacement while locating the shaft at the front bearing would create an artificial thermal offset and consume one lip gap. The selected centered representative is checked against both deformed core-to-lip clearances; it is not an assumed bonded axial interface.

All reported design stresses below are unaveraged element integration-point values. Smoothed nodal stresses mix material interfaces and are retained for visualization only. Composite von Mises stress is not a composite failure criterion, and magnet tensile strength cannot be assessed from steel yield stress.

The final 2 MPa scalar pressure is approximately 52% above the largest sampled outer-gap $|B|^2/(2\mu_0)$ value in the new d-axis fault case. It is a deliberately increased screening load, not a direct mapped Maxwell traction analysis or a mesh-converged local peak bound. Higher sharp-edge fields occur inside the inner bond/segment seam. Physical edge radii and a resolved local electromagnetic/contact submodel are required for local strength qualification.

Normal pressure uses native element-face follower loading, updating surface normals and area with deformation. Tangential nodal loads have 13.416 N·m nominal moment in the undeformed geometry; the actual applied and reaction moments are evaluated in the deformed geometry. Fixed global radial nodal forces were excluded because they introduce an artificial drag moment when the rotor turns.

19A. Structural constitutive inputs and constraints

Isotropic material	E [Pa]	Poisson ν	Density [kg/m³]	CTE [µm/m·K]
Steel	2.10×10^{11}	0.3	7650	12
Magnet	1.50×10^{11}	0.24	8300	6
Linear comparison bond	2.60×10^{6}	0.3	1200	80

Sleeve cylindrical material axis	Declared engineering constant
Axis 1 / 2 / 3	Radial / hoop / axial
E1 / E2 / E3	10 / 100 / 10 GPa
ν12 / ν13 / ν23	0.025 / 0.25 / 0.25; reciprocal values follow elastic symmetry
G12 / G13 / G23	5 / 3 / 5 GPa
Thermal expansion α1 / α2 / α3	25 / 0 / 0 µm/m·K
Density	1600 kg/m³

Nonlinear bond input	Value
Compressible neo-Hookean C10	0.500 MPa
D1	9.230769×10^{-7} Pa⁻¹
Equivalent reference G / K	1.000 / 2.166667 MPa
Stress-free temperature	20°C
Bond CTE	80 µm/m·K

Each radial bearing contributes 5 MN/m through an on-axis reference point coupled to the affine mean journal translation. The averaging annihilates rigid shaft rotation. Mean output-tip rotation fixes phase; mean front-journal axial displacement fixes the axial reference. Sleeve rotation is stabilized by a weak 0.01 N·m/rad numerical spring; its actual torque must remain below 0.001 N·m. The centered axial sleeve reference follows the core midplane. Contact is frictionless with a 10^{12} Pa/m normal penalty. No adhesive damage, creep, composite interaction failure, rolling bearing contact or fit interference is included.

Rotor laminations are represented mechanically by isotropic steel in this screening model. Interlaminar slip, stack preload, selected shaft fit/joint capacity and actual orthotropic stack mechanics remain to be assigned. The 1,890 rpm case is a mechanical stress screen; it does not assert available loaded torque at that speed on the 48 V inverter.

The corrected load path is verified independently. A pure 13.416 N·m torque solve gives an output reaction of −13.416000 N·m. Away from interfaces, the Ø12 shaft shear is 39.614 MPa versus the analytical $16T/(\pi d^3)$ = 39.541 MPa, a 0.184% difference. The final hot-contact output reaction is also checked against the full applied torque before acceptance.

20. Stress, sleeve interface and the design revision

Model	Material	Hoop max [MPa]	Tensile principal max	Shear max	VM p99 [MPa]
A bonded, corrected supports	wrap	435.771	761.158	512.778	486.362
A bonded, corrected supports	bond	-0.037	0.026	1.936	1.055
A bonded, corrected supports	magnet	130.953	258.398	153.983	186.584
A bonded, corrected supports	steel	112.012	130.557	238.589	99.662
B sliding, corrected supports	wrap	215.875	217.258	129.077	169.968
B sliding, corrected supports	bond	3.130	4.357	0.681	0.651
B sliding, corrected supports	magnet	2.828	7.645	6.176	5.645
B sliding, corrected supports	steel	48.991	87.329	73.206	77.497

With the corrected supports, the A bonded linear-tetrahedron comparison gives approximately 435.8 MPa peak sleeve hoop stress and 258.4 MPa peak magnet tensile principal stress. Its sleeve hoop requirement alone is approximately 871.5 MPa at factor two. These are comparison requirements, not strength passes. B introduces axial clearance and a sliding interface to release the original end-face restraint.

Earlier supports grounded the sleeve angle and distributed Cartesian bearing-node motions, allowing unintended external torque reactions. Tying sleeve rotation rigidly to the core also produced an artificial internal torque through faceted contact. Those results are excluded from final strength evidence. The final mean-bearing/free-sleeve model checks both output reaction and the weak numerical spring torque. Sharp interface peaks and contact penalty/mesh effects still require qualification; historical P1/P2 agreement under the earlier supports is not treated as convergence of this contact solution.

A and B are different screening scenarios: B changes the pressure, contact law, axial capture and displacement basis. Their stress difference cannot be attributed solely to the added lip clearance. The A comparison provides a restraint/mesh warning; the final B numbers apply to its explicitly stated 2 MPa centered sliding model.

Revision B completed contact check	Result
Interface	Frictionless node-to-surface, small sliding; thermal preheat then fault load
Applied final screening pressure	2.000 MPa; A comparison used 0.591525 MPa
Normal penalty	1×10^{12} Pa/m
Peak / p99 penetration	13.466 / 11.928 µm
Peak / p99 contact pressure	10.848 / 9.389 MPa
Outer radial growth	49.215 µm
Required sleeve hot hoop strength, factor two	431.750 MPa

The final contact model uses linear tetrahedra and a finite-strain soft-bond law. The larger sliding surface formulation and earlier time-limited runs were excluded. A small-sliding contact model is appropriate only while relative motion remains small compared with its facets; penalty penetration and mesh/facet effects are numerical uncertainty. Composite interaction failure, adhesive creep/fatigue and brittle magnet strength still require selected material data.

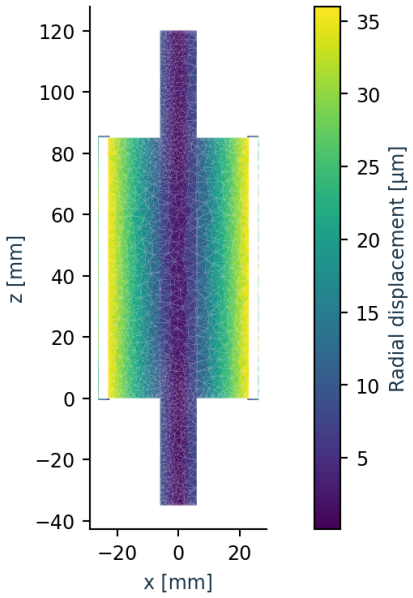
The numerical rotational spring carries 0.0003158 N·m; output reaction is -13.502385 N·m against 13.416 N·m nominal tangential loading. The proxy coordinate is $q = 0.02625\ \theta_s$ in metres, with translational spring stiffness $0.01/(0.02625^2) = 14.51247$ N/m. This removes the floating mode with negligible torque rather than rigidly holding the sleeve angle.

The actual current-geometry applied moment is 13.502634 N·m, including the follower-pressure contribution 0.060383 N·m. The complete deformed-geometry balance residual is 0.000067 N·m. Structural p99 values are percentiles by integration-point count; they are not volume-weighted percentiles.

Sleeve material-axis stress	Peak [MPa]	p99 [MPa]	Factor-two peak allowable [MPa]
radial tension	65.261	26.645	130.523
radial compression	106.945	27.626	213.889
hoop tension	215.875	164.385	431.750
hoop compression	6.914	0.000	13.829
axial tension	41.446	10.270	82.893
axial compression	54.257	24.742	108.515
shear radial hoop	19.790	9.742	39.580
shear radial axial	22.637	10.730	45.273
shear hoop axial	13.285	3.798	26.570

Principal maximum shear is an invariant on rotated planes; it is not automatically the laminate interlaminar shear stress. The table above gives actual radial/hoop/axial normal and shear components for comparison to selected composite allowables. A combined failure criterion and qualified laminate layup are still required.

The conservative deformed front/rear axial capture clearances are 30.319 / 30.585 µm. These use the extrema of the relevant end faces, so a positive value confirms that the assumed open axial lip interfaces remain open in the analysed centered hot load case. Manufacturing and assembly errors are not included.



21. Air-gap budget, bearing support and rotor modes

Clearance contribution	Value / calculation
Cold radial free gap	250 µm
A corrected-support hot-fault outer growth	41.556 µm
A residual gap before tolerance/runout	208.444 µm
Declared residual-gap target	100 µm
A combined tolerance/runout/deflection budget	108.444 µm
Static radial support compliance	100 N / 10 MN/m = 10 µm, plus shaft bending
First corrected-support unprestressed mode	416.0083 Hz
Equivalent 1× mode speed	24960.5 rpm; not an installed critical-speed map

The modal comparison uses the A bonded rotor and the corrected mean-bearing springs; it is not a prestressed solution of the final B sliding contact state. The first two modes are 416.008/416.013 Hz and the next is 525.304 Hz. Shaft rotation at 1,890 rpm is 31.5 Hz, but electromagnetic harmonics can excite much higher frequencies. Comparing only 1× shaft frequency with the first mode would miss possible coincidences.

Forcing order × shaft speed	Speed at first-mode coincidence	Interpretation
1	24960.5 rpm	Unbalance crossing above speed range
10	2496.0 rpm	2× electrical-frequency order; above maximum
12	2080.0 rpm	Slot-passing crossing above screening maximum
20	1248.0 rpm	Harmonic crossing inside screening range
30	832.0 rpm	Harmonic crossing inside screening range
40	624.0 rpm	Possible crossing near the 630 rpm duty

These are frequency coincidences, not predictions of vibration amplitude. Force spatial order, modal projection, damping, real bearing/housing compliance and the actual spectrum determine whether a mode is excited. Bearing speed dependence, preload, gyroscopic terms and installed support dynamics are absent. The results therefore identify a follow-up forced-response/installed-support requirement and do not establish a resonance-free operating range.

Minimum running clearance must be checked with the selected sleeve/interface result, actual measured runout, rotor/stator diameter tolerances, bearing clearance and magnetic pull. A nominal 0.25 mm gap is not a tolerance-qualified running gap. The CAD fit does not allocate these unknown errors automatically.

For the completed revision B contact scenario, the calculated outer radial growth is 49.215 µm. The nominal residual gap is therefore 200.785 µm; retaining a 100 µm residual target leaves 100.785 µm for the combined unmodelled tolerance, runout and deflection budget. Stator thermal expansion is omitted from this conservative clearance subtraction.

21A. Nominal assembly mass and rotor inertia

Material class	Nominal mass [kg]
steel	3.690248
winding	0.448372
insulation	0.015520
magnet	0.260258
bond	0.000588
wrap	0.005404
aluminum	1.096600
bearing_equivalent	0.069115

Quantity	Value
Nominal complete CAD assembly mass	5.586106 kg
Active rotating FE body mass	1.357110 kg
Active rotating polar inertia	0.0004340688 kg·m²

The nominal assembly includes housing, end caps, fastener envelopes, equivalent bearings, insulation and impregnated winding. Laminated iron mass uses 95% steel fill. Winding density follows the actual copper/insulation/resin area fractions; bearing density is an equivalent 5000 kg/m³. Masses come from independent OpenCascade integration of the delivered revision B solids. The rotating inertia includes the shaft, rotor iron, magnets, bond and sleeve; it excludes the rotating fraction of bearings, the coupling and the actuator.

22. Design decision and remaining qualification inputs

The study supplies a concrete CAD/FEA candidate and executed numerical evidence. A hardware strength or irreversible-demagnetization pass requires the actual supplier curves and allowables. The sleeve attachment, axial capture clearance and adhesive law are design variables; treating them as arbitrary bonded interfaces changes the result materially.

Gate	Status	Evidence / implication
Nominal CAD geometry	PASS	143 valid solids and STEP round trip; Ø112 housing, Ø12×155 shaft; revision B axial sleeve clearance 0.15 mm each end.
Native Pyleecan/Elmer coupling	PASS	24 positions; mean torque differs from native FEMM by 0.290%.
3D torque numerical checks	PASS	Nonlinear coarse/fine torque change 0.635%; component, discrete air Maxwell tensor and finite-rotation energy checked.
220 N·m mean duty	CONDITIONAL PASS	At 12.2 A/120°C magnets, mean shaft 4.6273 N·m after 18 W allowance; actuator 233.21 N·m with assumed 63:1/80%; margin 6.01%. Generic materials and loss allowance apply.
42 V nominal-duty voltage	CONDITIONAL PASS	Flux-derived sampled-harmonic phase-voltage circle 16.225 V peak versus 21.824 V reserved ceiling; switching and higher harmonics unresolved.
Torque-trough control headroom	CONDITIONAL PASS	Constant 12.2 A sampled minimum misses the shaft target; 13 A/150°C magnets at the sampled trough gives 4.4792 N·m shaft versus 4.3651. Ripple compensation, tracking and continuum extrema remain to qualify.
12.2 A hot holding	PASS	Refined winding 92.870°C; 70°C sink, low resin conductivity, reduced interface conductance, 30° holding angle.
Thermal joint retention	UNQUALIFIED	The 1000–5000 W/m²K contact conductance must be maintained by an engineered fit or bond. With generic Al/steel CTE, common 50 K heating opens a nominally zero-clearance Ø96 interface by 26.4 µm radially unless retained.
18.5 A continuous hot holding	CONDITIONAL PASS	Refined winding 126.387°C, only 3.613°C below 130°C; requires the stated cooling and impregnation.
22 A continuous hot holding	FAIL	Winding 154.732°C exceeds 130°C. Do not assign 22 A continuous holding.
22 A for 5 seconds	CONDITIONAL PASS	Refined winding 103.266°C from the 12.2 A hot holding field; fixed 130°C-resistance source and stated heat capacities/cooling.

22A. Mechanical and material qualification

Gate	Status	Evidence / implication
Rotor contact calculation	NUMERICALLY COMPLETED	Thermal preheat then full 1890 rpm/13.416 N·m/2 MPa/150°C case; finite-strain bond, frictionless small sliding. Uniform pressure is a screening envelope, not mapped local traction.
Structural torque load path	PASS	Current-geometry applied 13.502634 N·m, output reaction -13.502385 N·m, weak reference 0.000316 N·m; residual 0.000067 N·m. Independent shaft torsion error 0.184%.
Hot axial sleeve capture	NOMINAL PASS	Deformed front/rear lip gaps 30.3/30.6 µm with the centered floating sleeve; manufacturing tolerances excluded.
Sleeve, bond and magnet strength	UNQUALIFIED	Sleeve factor-two peak requirements: hoop 431.8, radial tension 130.5, axial tension 82.9 MPa. Transverse/shear peaks and sharp corners require checked layup, mesh/penalty sensitivity and actual hot allowables; adhesive creep and brittle-magnet data absent.
Irreversible demagnetization	UNQUALIFIED	3D −55.27 A d-axis/150°C reverse H max 712 kA/m; inherited cold fault requirement 852 kA/m. Supplier intrinsic curves/knee are absent.
Running gap and installed bearings	CONDITIONAL	Nominal residual gap 200.8 µm in the B hot fault case, before tolerances/runout. Generic bearing stiffness/conductance; fits, preload and supplier ratings not assigned.
Installed vibration / forced response	UNQUALIFIED	Corrected-support A bonded first mode 416.008 Hz; 40× shaft-order coincidence is 624.0 rpm, near 630 rpm duty. Actual excitation projection, damping and installed bearing/housing dynamics are required; coincidence alone is not amplitude.
Optional 400 N·m drive point	FAIL	Inherited angle-resolved 42 V reserved-voltage check failed; no pass is asserted by the new 22 A thermal or torque cases.

Retain the full-period native electromagnetic maps and controller voltage limit when comparing to the actuator. Gearbox efficiency, actuator shock loads and installed thermal boundaries are assumptions inherited from the requirement; the actuator itself is not analysed by the Pyleecan motor model.

23. CAD and Ansys transfer

Artifact	Units / contents
Motor_Gmsh_Elmer_Candidate_mm.step	Millimetres; named nominal 143-solid assembly, independent round-trip check
Motor_Gmsh_Elmer_Candidate.FCStd	FreeCAD document with per-part names and analysis material labels
Magnetic Gmsh .msh / Abaqus .inp	Metres; axial half-domain, labelled field regions and boundaries
Thermal Gmsh .msh / .inp	Metres; actual lined coil/end-turn geometry and declared equivalent contact region
Rotor CalculiX .inp	SI m, N, Pa, kg, s; explicit loads, materials, supports and contact definitions
Reproduction source / case records	Project adapters, F90 material/gauge callbacks, solver inputs and compact result tables

STEP imports in millimetres. Solver meshes/decks use metres: do not apply the STEP scale to an already-SI mesh. Check the imported shaft length 155 mm and housing diameter 112 mm. Bearings and fasteners remain explicit dimensional envelopes. Closed coil solids represent stranded winding regions; do not assign them solid-copper eddy-current conductivity in a magnetic solver.

When recreating the magnetic half-model, preserve Bnormal = 0 at the stack midplane, the stated far-domain planes, current directions, 12 turns per coil, magnet radial orientation, coil/end-turn geometry and lamination tensor. In a full model, remove the symmetry multiplier. For thermal analysis, preserve complete barrel cooling and resolve the physical liner or a documented equivalent resistance.

No Ansys solve is claimed. The open-source model inputs and results are provided so that the user can independently recreate the same physics, then refine contacts, materials, mesh and installed conditions in Ansys or another solver.

The optional model archive also contains moving13_hot70_temperature_full_SI.csv: a 349,587-point full-motor temperature cloud obtained by mirroring the executed axial-half 13 A moving solution. Coordinates are metres and temperature is °C. It supports independent interpolation/comparison; the structural screen in this report still uses uniform 150°C, not a claimed mapped-temperature structural run.

24. Numerical provenance and result boundaries

Software / input	Recorded basis
Pyleecan	1.5.2; installed Python source integrity checked against distribution RECORD
Gmsh	4.13; OpenCascade geometry, conformal meshes
Elmer	Official release 26.2 source; local serial and MPI/HYPRE builds
Elmer CHOLMOD integration	Recorded 64-bit-index type correction; reference scalar solve versus UMFPACK
Elmer nonlinear tensor Newton	Recorded differentiation of complete H(B) vector; used by the nonlinear 3D HPC cases
Project extensions	Boundary-rooted tree gauge; motor-specific Pyleecan adapter; lamination and curved-winding material callbacks
CalculiX	2.20; linear/quadratic tetrahedra, orthotropic sleeve and nonlinear contact
Independent checks	scikit-fem/SciPy magnetic and conducting-potential solves; FreeCAD/OpenCascade geometry readback
Elmer source archive SHA-256	7addfb102d735929a40929be08be0c16703d734498d33bc70c96e7d526fe249c

All six selected upstream reference tests passed after the nonlinear correction: HeatAniso, mgdyn_anisotropic_rel, mgdyn_anisotropic_rel2, mgdyn_bh_relfun, mgdyn_hypre_ams and mgdyn_steady_2ndorder. The CHOLMOD scalar comparison to UMFPACK gives a 2.52×10^{-10} relative field difference. The project also checks finite-rotation energy torque, discrete Maxwell torque, native API results, wire packing, mesh/time refinement and mechanical load balance. Rejected and time-limited runs are excluded from accepted-result tables.

The recorded Elmer source corrections and project callbacks are required to reproduce these particular nonlinear 3D runs; results are not presented as an unmodified stock MagElmer workflow. Build recipes and hashes are supplied, although installation into a fresh empty host has not been tested. There is no claim of an Ansys solve or hardware correlation.

This report does not claim that every analysis Elmer or Pyleecan can perform has been run. There is no finite exhaustive software checklist that proves a motor correct. Its conclusions apply to the declared geometry, loads, material assumptions and numerical checks. Supplier-based demagnetization, composite/adhesive failure, manufacturing fits and physical test correlation remain qualification inputs.

Appendix A. Complete solid inventory and CAD bounds

Nominal axis-aligned CAD bounds in millimetres. Kernel bounding boxes for curved coils include a small numerical tolerance; use the exact swept geometry when constructing end turns. All 143 solids passed independent BREP validity checking and STEP round-trip count checks for the delivered revision.

Part name	Material class	Minimum x,y,z [mm]	Maximum x,y,z [mm]
StatorIron	steel	-48.000, -48.000, -0.000	48.000, 48.000, 85.000
RotorIron	steel	-23.000, -23.000, -0.000	23.000, 23.000, 85.000
Shaft	steel	-6.000, -6.000, -35.000	6.000, 6.000, 120.000
Coil01_A	winding	27.693, -10.659, -10.659	40.995, 10.659, 95.659
Coil02_A	winding	20.293, 7.448, -10.659	39.885, 29.178, 95.659
Coil03_B	winding	7.448, 20.293, -10.659	29.178, 39.885, 95.659
Coil04_B	winding	-10.659, 27.693, -10.659	10.659, 40.995, 95.659
Coil05_C	winding	-29.178, 20.293, -10.659	-7.448, 39.885, 95.659
Coil06_C	winding	-39.885, 7.448, -10.659	-20.293, 29.178, 95.659
Coil07_A	winding	-40.995, -10.659, -10.659	-27.693, 10.659, 95.659
Coil08_A	winding	-39.885, -29.178, -10.659	-20.293, -7.448, 95.659
Coil09_B	winding	-29.178, -39.885, -10.659	-7.448, -20.293, 95.659
Coil10_B	winding	-10.659, -40.995, -10.659	10.659, -27.693, 95.659
Coil11_C	winding	7.448, -39.885, -10.659	29.178, -20.293, 95.659
Coil12_C	winding	20.293, -29.178, -10.659	39.885, -7.448, 95.659
LinerSeparator01	insulation	27.252, 6.389, -0.000	41.232, 15.083, 85.000
SlotWedge01	insulation	26.286, 4.474, -0.000	28.287, 10.269, 85.000
LinerSeparator02	insulation	19.432, 19.432, -0.000	32.514, 32.514, 85.000
SlotWedge02	insulation	18.018, 18.018, -0.000	22.261, 22.261, 85.000
LinerSeparator03	insulation	6.389, 27.252, -0.000	15.083, 41.232, 85.000
SlotWedge03	insulation	4.474, 26.286, -0.000	10.269, 28.287, 85.000
LinerSeparator04	insulation	-15.083, 27.252, -0.000	-6.389, 41.232, 85.000
SlotWedge04	insulation	-10.269, 26.286, -0.000	-4.474, 28.287, 85.000
LinerSeparator05	insulation	-32.514, 19.432, -0.000	-19.432, 32.514, 85.000
SlotWedge05	insulation	-22.261, 18.018, -0.000	-18.018, 22.261, 85.000
LinerSeparator06	insulation	-41.232, 6.389, -0.000	-27.252, 15.083, 85.000
SlotWedge06	insulation	-28.287, 4.474, -0.000	-26.286, 10.269, 85.000
LinerSeparator07	insulation	-41.232, -15.083, -0.000	-27.252, -6.389, 85.000
SlotWedge07	insulation	-28.287, -10.269, -0.000	-26.286, -4.474, 85.000
LinerSeparator08	insulation	-32.514, -32.514, -0.000	-19.432, -19.432, 85.000
SlotWedge08	insulation	-22.261, -22.261, -0.000	-18.018, -18.018, 85.000
LinerSeparator09	insulation	-15.083, -41.232, -0.000	-6.389, -27.252, 85.000
SlotWedge09	insulation	-10.269, -28.287, -0.000	-4.474, -26.286, 85.000

Part name	Material class	Minimum x,y,z [mm]	Maximum x,y,z [mm]
LinerSeparator10	insulation	6.389, -41.232, -0.000	15.083, -27.252, 85.000
SlotWedge10	insulation	4.474, -28.287, -0.000	10.269, -26.286, 85.000
LinerSeparator11	insulation	19.432, -32.514, -0.000	32.514, -19.432, 85.000
SlotWedge11	insulation	18.018, -22.261, -0.000	22.261, -18.018, 85.000
LinerSeparator12	insulation	27.252, -15.083, -0.000	41.232, -6.389, 85.000
SlotWedge12	insulation	26.286, -10.269, -0.000	28.287, -4.474, 85.000
Magnet01_Segment1	magnet	19.462, 1.447, -0.000	25.999, 13.958, 21.175
Bond01_Segment1	bond	19.419, 1.444, -0.000	23.005, 12.351, 21.175
Magnet01_Segment2	magnet	19.462, 1.447, 21.275	25.999, 13.958, 42.450
Bond01_Segment2	bond	19.419, 1.444, 21.275	23.005, 12.351, 42.450
Magnet01_Segment3	magnet	19.462, 1.447, 42.550	25.999, 13.958, 63.725
Bond01_Segment3	bond	19.419, 1.444, 42.550	23.005, 12.351, 63.725
Magnet01_Segment4	magnet	19.462, 1.447, 63.825	25.999, 13.958, 85.000
Bond01_Segment4	bond	19.419, 1.444, 63.825	23.005, 12.351, 85.000
Magnet02_Segment1	magnet	8.485, 14.693, -0.000	20.072, 24.221, 21.175
Bond02_Segment1	bond	8.467, 14.661, -0.000	17.760, 21.431, 21.175
Magnet02_Segment2	magnet	8.485, 14.693, 21.275	20.072, 24.221, 42.450
Bond02_Segment2	bond	8.467, 14.661, 21.275	17.760, 21.431, 42.450
Magnet02_Segment3	magnet	8.485, 14.693, 42.550	20.072, 24.221, 63.725
Bond02_Segment3	bond	8.467, 14.661, 42.550	17.760, 21.431, 63.725
Magnet02_Segment4	magnet	8.485, 14.693, 63.825	20.072, 24.221, 85.000
Bond02_Segment4	bond	8.467, 14.661, 63.825	17.760, 21.431, 85.000
Magnet03_Segment1	magnet	-6.478, 22.326, -0.000	6.478, 26.050, 21.175
Bond03_Segment1	bond	-5.732, 22.277, -0.000	5.732, 23.050, 21.175
Magnet03_Segment2	magnet	-6.478, 22.326, 21.275	6.478, 26.050, 42.450
Bond03_Segment2	bond	-5.732, 22.277, 21.275	5.732, 23.050, 42.450
Magnet03_Segment3	magnet	-6.478, 22.326, 42.550	6.478, 26.050, 63.725
Bond03_Segment3	bond	-5.732, 22.277, 42.550	5.732, 23.050, 63.725
Magnet03_Segment4	magnet	-6.478, 22.326, 63.825	6.478, 26.050, 85.000
Bond03_Segment4	bond	-5.732, 22.277, 63.825	5.732, 23.050, 85.000
Magnet04_Segment1	magnet	-20.072, 14.693, -0.000	-8.485, 24.221, 21.175
Bond04_Segment1	bond	-17.760, 14.661, -0.000	-8.467, 21.431, 21.175
Magnet04_Segment2	magnet	-20.072, 14.693, 21.275	-8.485, 24.221, 42.450
Bond04_Segment2	bond	-17.760, 14.661, 21.275	-8.467, 21.431, 42.450
Magnet04_Segment3	magnet	-20.072, 14.693, 42.550	-8.485, 24.221, 63.725
Bond04_Segment3	bond	-17.760, 14.661, 42.550	-8.467, 21.431, 63.725
Magnet04_Segment4	magnet	-20.072, 14.693, 63.825	-8.485, 24.221, 85.000

Part name	Material class	Minimum x,y,z [mm]	Maximum x,y,z [mm]
Bond04_Segment4	bond	-17.760, 14.661, 63.825	-8.467, 21.431, 85.000
Magnet05_Segment1	magnet	-25.999, 1.447, -0.000	-19.462, 13.958, 21.175
Bond05_Segment1	bond	-23.005, 1.444, -0.000	-19.419, 12.351, 21.175
Magnet05_Segment2	magnet	-25.999, 1.447, 21.275	-19.462, 13.958, 42.450
Bond05_Segment2	bond	-23.005, 1.444, 21.275	-19.419, 12.351, 42.450
Magnet05_Segment3	magnet	-25.999, 1.447, 42.550	-19.462, 13.958, 63.725
Bond05_Segment3	bond	-23.005, 1.444, 42.550	-19.419, 12.351, 63.725
Magnet05_Segment4	magnet	-25.999, 1.447, 63.825	-19.462, 13.958, 85.000
Bond05_Segment4	bond	-23.005, 1.444, 63.825	-19.419, 12.351, 85.000
Magnet06_Segment1	magnet	-25.999, -13.958, -0.000	-19.462, -1.447, 21.175
Bond06_Segment1	bond	-23.005, -12.351, -0.000	-19.419, -1.444, 21.175
Magnet06_Segment2	magnet	-25.999, -13.958, 21.275	-19.462, -1.447, 42.450
Bond06_Segment2	bond	-23.005, -12.351, 21.275	-19.419, -1.444, 42.450
Magnet06_Segment3	magnet	-25.999, -13.958, 42.550	-19.462, -1.447, 63.725
Bond06_Segment3	bond	-23.005, -12.351, 42.550	-19.419, -1.444, 63.725
Magnet06_Segment4	magnet	-25.999, -13.958, 63.825	-19.462, -1.447, 85.000
Bond06_Segment4	bond	-23.005, -12.351, 63.825	-19.419, -1.444, 85.000
Magnet07_Segment1	magnet	-20.072, -24.221, -0.000	-8.485, -14.693, 21.175
Bond07_Segment1	bond	-17.760, -21.431, -0.000	-8.467, -14.661, 21.175
Magnet07_Segment2	magnet	-20.072, -24.221, 21.275	-8.485, -14.693, 42.450
Bond07_Segment2	bond	-17.760, -21.431, 21.275	-8.467, -14.661, 42.450
Magnet07_Segment3	magnet	-20.072, -24.221, 42.550	-8.485, -14.693, 63.725
Bond07_Segment3	bond	-17.760, -21.431, 42.550	-8.467, -14.661, 63.725
Magnet07_Segment4	magnet	-20.072, -24.221, 63.825	-8.485, -14.693, 85.000
Bond07_Segment4	bond	-17.760, -21.431, 63.825	-8.467, -14.661, 85.000
Magnet08_Segment1	magnet	-6.478, -26.050, -0.000	6.478, -22.326, 21.175
Bond08_Segment1	bond	-5.732, -23.050, -0.000	5.732, -22.277, 21.175
Magnet08_Segment2	magnet	-6.478, -26.050, 21.275	6.478, -22.326, 42.450
Bond08_Segment2	bond	-5.732, -23.050, 21.275	5.732, -22.277, 42.450
Magnet08_Segment3	magnet	-6.478, -26.050, 42.550	6.478, -22.326, 63.725
Bond08_Segment3	bond	-5.732, -23.050, 42.550	5.732, -22.277, 63.725
Magnet08_Segment4	magnet	-6.478, -26.050, 63.825	6.478, -22.326, 85.000
Bond08_Segment4	bond	-5.732, -23.050, 63.825	5.732, -22.277, 85.000
Magnet09_Segment1	magnet	8.485, -24.221, -0.000	20.072, -14.693, 21.175
Bond09_Segment1	bond	8.467, -21.431, -0.000	17.760, -14.661, 21.175
Magnet09_Segment2	magnet	8.485, -24.221, 21.275	20.072, -14.693, 42.450
Bond09_Segment2	bond	8.467, -21.431, 21.275	17.760, -14.661, 42.450

Part name	Material class	Minimum x,y,z [mm]	Maximum x,y,z [mm]
Magnet09_Segment3	magnet	8.485, -24.221, 42.550	20.072, -14.693, 63.725
Bond09_Segment3	bond	8.467, -21.431, 42.550	17.760, -14.661, 63.725
Magnet09_Segment4	magnet	8.485, -24.221, 63.825	20.072, -14.693, 85.000
Bond09_Segment4	bond	8.467, -21.431, 63.825	17.760, -14.661, 85.000
Magnet10_Segment1	magnet	19.462, -13.958, -0.000	25.999, -1.447, 21.175
Bond10_Segment1	bond	19.419, -12.351, -0.000	23.005, -1.444, 21.175
Magnet10_Segment2	magnet	19.462, -13.958, 21.275	25.999, -1.447, 42.450
Bond10_Segment2	bond	19.419, -12.351, 21.275	23.005, -1.444, 42.450
Magnet10_Segment3	magnet	19.462, -13.958, 42.550	25.999, -1.447, 63.725
Bond10_Segment3	bond	19.419, -12.351, 42.550	23.005, -1.444, 63.725
Magnet10_Segment4	magnet	19.462, -13.958, 63.825	25.999, -1.447, 85.000
Bond10_Segment4	bond	19.419, -12.351, 63.825	23.005, -1.444, 85.000
RetainingWrap	wrap	-26.250, -26.250, -0.150	26.250, 26.250, 85.150
FrontRetentionLip	wrap	-26.250, -26.250, -0.650	26.250, 26.250, -0.150
RearRetentionLip	wrap	-26.250, -26.250, 85.150	26.250, 26.250, 85.650
HousingBarrel	aluminum	-56.000, -56.000, -0.000	56.000, 56.000, 85.000
EndCap_left	aluminum	-56.000, -56.000, -20.000	56.000, 56.000, 0.000
EndCap_right	aluminum	-56.000, -56.000, 85.000	56.000, 56.000, 105.000
FrontBearing	bearing_equivalent	-16.000, -16.000, -19.000	16.000, 16.000, -9.000
RearBearing	bearing_equivalent	-16.000, -16.000, 94.000	16.000, 16.000, 104.000
FrontInnerCollar	steel	-9.000, -9.000, -9.000	9.000, 9.000, -5.000
RearInnerCollar	steel	-9.000, -9.000, 90.000	9.000, 9.000, 94.000
FrontLocknut	steel	-9.500, -9.500, -22.000	9.500, 9.500, -19.000
RearLocknut	steel	-9.500, -9.500, 104.000	9.500, 9.500, 107.000
FrontM4Bolt1	steel	41.533, 22.500, -24.000	48.533, 29.500, 10.000
RearM4Bolt1	steel	41.533, 22.500, 75.000	48.533, 29.500, 109.000
FrontM4Bolt2	steel	-3.500, 48.500, -24.000	3.500, 55.500, 10.000
RearM4Bolt2	steel	-3.500, 48.500, 75.000	3.500, 55.500, 109.000
FrontM4Bolt3	steel	-48.533, 22.500, -24.000	-41.533, 29.500, 10.000
RearM4Bolt3	steel	-48.533, 22.500, 75.000	-41.533, 29.500, 109.000
FrontM4Bolt4	steel	-48.533, -29.500, -24.000	-41.533, -22.500, 10.000
RearM4Bolt4	steel	-48.533, -29.500, 75.000	-41.533, -22.500, 109.000
FrontM4Bolt5	steel	-3.500, -55.500, -24.000	3.500, -48.500, 10.000
RearM4Bolt5	steel	-3.500, -55.500, 75.000	3.500, -48.500, 109.000
FrontM4Bolt6	steel	41.533, -29.500, -24.000	48.533, -22.500, 10.000
RearM4Bolt6	steel	41.533, -29.500, 75.000	48.533, -22.500, 109.000

Appendix B. Magnetic material curve and lamination law

B in steel [T]	H [A/m]	Basis
0.000	0.000	Bundled M270-35A
0.100	30.000	Bundled M270-35A
0.200	39.600	Bundled M270-35A
0.300	46.000	Bundled M270-35A
0.400	52.000	Bundled M270-35A
0.500	58.200	Bundled M270-35A
0.600	65.200	Bundled M270-35A
0.700	73.300	Bundled M270-35A
0.800	83.100	Bundled M270-35A
0.900	95.500	Bundled M270-35A
1.000	112.000	Bundled M270-35A
1.100	136.000	Bundled M270-35A
1.200	178.000	Bundled M270-35A
1.300	272.000	Bundled M270-35A
1.400	596.000	Bundled M270-35A
1.500	1700.000	Bundled M270-35A
1.600	3880.000	Bundled M270-35A
1.700	7160.000	Bundled M270-35A
1.800	11600.000	Bundled M270-35A
2.000	20480.000	Linear high-field extrapolation
2.500	42680.000	Linear high-field extrapolation
3.000	64880.000	Linear high-field extrapolation
4.000	109280.000	Linear high-field extrapolation

The tabulated isotropic steel curve is interpolated linearly in H(B). For fill kf = 0.95, solve $\nu_{Fe} = H(|B_{Fe}|)/|B_{Fe}|$ with $B_{Fe,parallel} = B_{parallel}/[k_f+(1-k_f)\mu_0\nu_{Fe}]$ and $B_{Fe,z} = B_z$. The homogenized reluctivity is $\nu_{parallel} = \nu_{Fe}/[k_f+(1-k_f)\mu_0\nu_{Fe}]$ and $\nu_z = k_f\nu_{Fe}+(1-k_f)/\mu_0$. This preserves a conservative nonlinear anisotropic laminate law, rather than simply multiplying a scalar permeability by fill.

The linear numerical check uses $\mu_{r,parallel} = 1500$ and $\mu_{r,z} = 19.75$. The PM recoil model uses $B_{r20} = 1.05$ T, $\alpha_{Br} = -0.0003\ K^{-1}$ and $\mu_r = 1.05$, with alternating radial magnetization. Its source magnetization is $B_r/(\mu_0\mu_r)$. The magnet's intrinsic demagnetization curve is not supplied by this recoil model.